

DIURNAL VARIATIONS IN RESPONSE OF GOLDEN SPINY MICE (*Acomys russatus*) TO NORADRENALINE INJECTION.

N. Kronfeld¹, N. Zisapel² and A. Haim³

¹Depts. of Zoology and ²Biochemistry, Tel-Aviv University, Tel-Aviv 69978, Israel. ³Dept. of Biology, University of Haifa at Oranim, Kiryat Tivon 36910, Israel.

Introduction

Non-shivering thermogenesis (NST) is an important mechanism for rapidly increasing heat production in winter-acclimatized, as well as in cold- and long-scotophase acclimated small mammals (1,2,3). The great advantage of regulatory NST as a thermogenic mechanism is that when a rodent is not exposed to low ambient temperatures, metabolism can be kept at a relatively low rate; during exposure to cold, heat production is increased within a relatively short period of time (4). Therefore, this mechanism enables rodents to conserve energy.

It is commonly considered that NST capacity for cold acclimated mammals is a function of body mass (2). However, recent studies indicate that capacity for NST is not only allometrically determined. It appears that natural selection acting through habitat or habits may play an important role. For example, it has been demonstrated that NST capacity in desert rodents that are characterized by low RMR's (resting metabolic rate) is relatively high (5).

The golden spiny mouse, *Acomys russatus* is distributed in Israel in the most extreme hot and arid rocky desert regions. Its activity pattern under natural conditions, in the presence of the common spiny mouse *A.cahirinus*, is that of diurnal species. Like other desert rodents, the golden spiny mice have lower RMR than expected (6). Elimination of the sympatric species, the common spiny mouse *A.cahirinus* from the common habitat results in nocturnal activity of *A. russatus* (7). Accordingly, when kept in captivity without individuals of *A. cahirinus*, the golden spiny mouse exhibits a nocturnal activity pattern (8). Rubal *et al.* (8) demonstrated that in the golden spiny mouse, the increase in oxygen consumption precedes the increase in body temperature by *ca.* 3 h. This result suggests that the golden spiny mouse behaves like the pouched mouse (*Saccostomus campestris*), in which the VO_2 rhythm rather than the overall thermal conductance rhythm was found to be the driving force in the T_b rhythm (9).

The low RMR of the golden spiny mouse, coupled with the daily temporal order in VO_2 and T_b rhythms, suggest that NST may be a major driving force in the T_b rhythm. In such case it should be assumed that NST would exhibit diurnal variations. In order to check this hypothesis we tested to what extent the thermogenic state of BAT (Brown adipose tissue) changes in response to NA (Noradrenaline) injection during the maximum and minimum phases of the T_b and VO_2 daily cycle in *A. russatus*.

Materials and methods

Golden spiny mice (8 females, mean body mass 77.8 ± 12.8 gr.) from a colony in the University of Haifa at Oranim (that were collected at Ein-Gedi on the Dead sea shore, and bred in Oranim) were used in the experiment. Prior to the experiment, the mice were individually caged with a photoperiod regime of 12L:12D (lights on 06:00 - 18:00 h) at $T_a = 28 \pm 1^\circ\text{C}$. Rat pellets and carrot were supplied *ad lib*.

The diurnal variations in the response of the golden spiny mice to noradrenaline (NA) injection were measured in one individual every 3 h in a metabolic chamber at $T_a = 31 \pm 1^\circ\text{C}$, i.e. their lower critical point (6). Oxygen consumption (VO_2) and body (rectal) temperature ($\text{VO}_{2\text{min}}$, $T_{b\text{min}}$) were recorded and immediately after that, NA was injected (1.5 mg/kg body mass), and VO_2 measurements were resumed for at least 30 min ($\text{VO}_{2\text{NA}}$). T_b was measured once again at the end of this period ($T_{b\text{NA}}$).

NST capacity was calculated from the net increase in O_2 consumption and in T_b due to the NA injections. O_2 consumption was measured in an open-circuit system (10,11) using flow of dried air at a rate of 480 ml/min. T_b was measured using a thermometer inserted 3 cm into the rectum of the tested mice.

Results

The results show clear and significant diurnal variation in the response of the animals VO_2 and T_b to NA injections. Maximal $T_{b\text{NA}}$ and $\text{VO}_{2\text{NA}}$ Values were recorded in the middle of the dark (at 24:00 h) and the light periods (at 12:00 h). The minimum values were recorded at the beginning and end of the dark period (6:00 and 18:00 h). The metabolic rate and body temperature measured in this and other studies, under the same conditions (8), indicate maximal T_b and VO_2 at the beginning and end of the dark period and minimal values in the middle of the dark and light periods (fig. 1,2). Here the ratio between $T_{b\text{NA}}/T_b$, $\text{VO}_{2\text{NA}}/\text{VO}_2$ and NST capacity are highest in the middle of the dark and light periods, and lowest at the beginning and end of the dark period (fig. 1,2).

Discussion

Our results clearly show daily variation in the response of the golden spiny mice to NA injections. This variation may be the result of a daily rhythm in the availability of NA receptors of BAT, or a result of the endogenous secretion of NA which occupies the available receptors. It is widely